SKILL BUILDERS · UNIT 7 · SCORE 3

Statistics: Practice Set

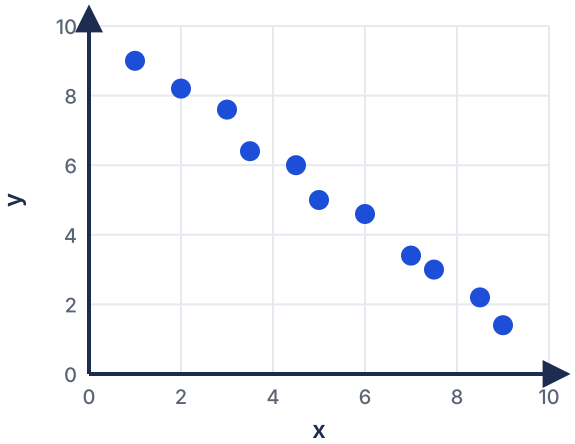
NC.8.SP.1–4

ANSWER KEY

Do one problem at a time. Show your steps. Circle the letter or write the number.

1 Score 3 NC.8.SP.1

What kind of correlation does this scatter plot show?



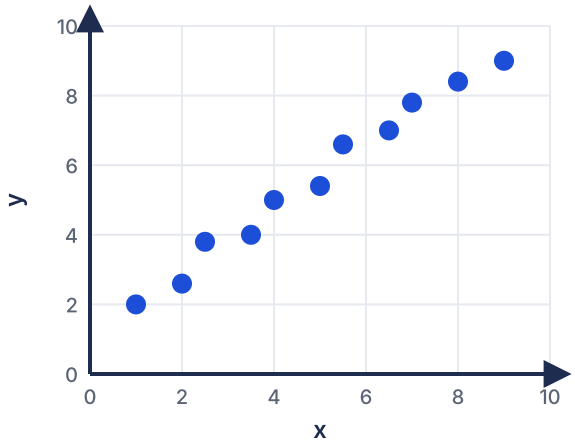
A **Negative correlation**

B Positive correlation

C No correlation

2 Score 3 NC.8.SP.1

What kind of correlation does this scatter plot show?



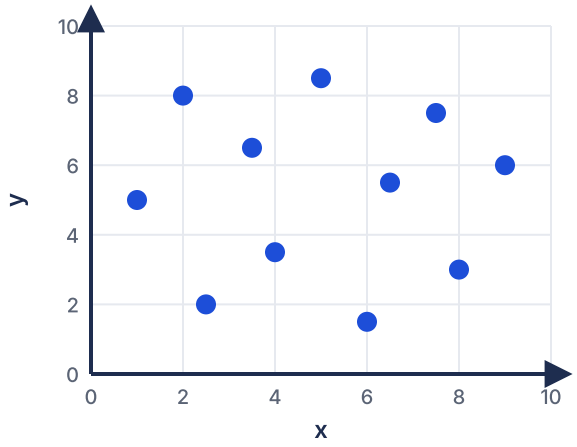
A **Positive correlation**

B Negative correlation

C No correlation

3 Score 3 NC.8.SP.1

What kind of correlation does this scatter plot show?



A **No correlation**

B Positive correlation

C Negative correlation

4 Score 3 NC.8.SP.1

A single dot far away from the rest of a scatter plot is called...

A **an outlier**

B a line of best fit

C a correlation

5 Score 3 NC.8.SP.3

Best fit: **y = 3x + 4**. Predict y when x = 5.

Answer: **19**

1 `y = 3(5) + 4`

2 `= 15 + 4 = 19`

6 Score 3 NC.8.SP.3

A best fit line for plant height is **y = 2x + 3** (x = weeks, y = cm). What does the slope 2 mean?

A The plant is 2 weeks old

B **The plant grows about 2 cm each week**

C The plant started at 2 cm

7 Score 3 NC.8.SP.2

Data go from x = 1 to x = 8. Using the line to predict y at x = 30 is...

A **extrapolation (riskier)**

B interpolation (safer)

8 Score 3 NC.8.SP.4

	Likes dogs	Does not	Total
Boys	12	8	**20**
Girls	15	5	**20**
Total	**27**	**13**	**40**

How many **girls** like dogs?

Answer: **15**

9 Score 3 NC.8.SP.4

	Likes dogs	Does not	Total
Boys	12	8	**20**
Girls	15	5	**20**
Total	**27**	**13**	**40**

Of the **boys**, what percent like dogs?

Answer: **60%**

1 `boys who like dogs: 12; all boys: 20`

2 `12 ÷ 20 = 0.60 = 60%`

10 Score 3 NC.8.SP.1

Find the mistake. A scatter plot shows that towns with more ice cream shops have more sunburns. Rio says: "Ice cream causes sunburns." What is wrong?

(A) Nothing. The data prove it.

(B) A correlation does not prove cause. Hot, sunny weather can cause both.

(C) The correlation must be negative.

(D) Sunburns cause ice cream shops.